

September dataset and experiment update

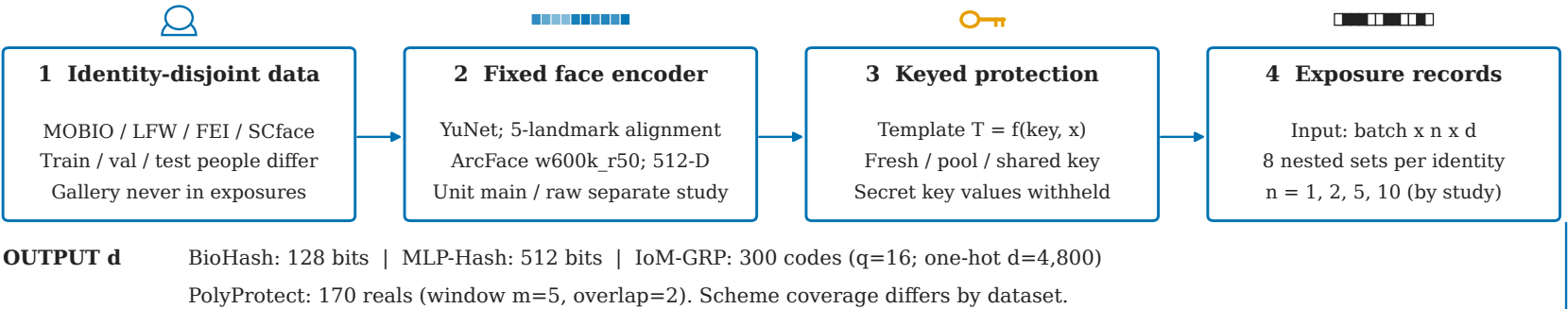
Identity-disjoint evaluation across session, pose and camera variation.

Dataset	Identities	Train / val / test	Valid / selected	Variation
MOBIO	150	90 / 30 / 30	1,799 / 1,800	Sessions
LFW	125	75 / 25 / 25	1,500 / 1,500	In-the-wild
FEI	200	120 / 40 / 40	2,378 / 2,400	Pose / expression
SCface	130	78 / 26 / 26	2,851 / 2,860	Camera / distance

SCface: mugshot-to-surveillance linkage under capture-domain shift.
Mugshot gallery / surveillance exposures; all 130 identities remain eligible after 9 detection failures.
Unprotected SCface top-1: 84.375% over 544 test probes; chance: $1/26 = 3.846\%$.
Evidence: 73 key-pool conditions / 12 studies; 72 one-seed model endpoints / 24 scheme pilot cells.
Earlier: 216 endpoints; 144 pool-study fits + 72 baseline evaluations; new: 672 MOBIO/SCface endpoints.

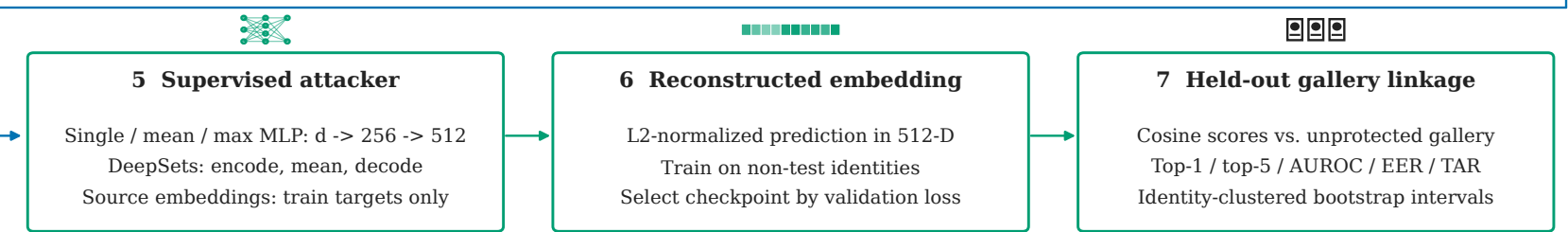
Experimental architecture

A DATA AND PROTECTION



B KEY-BLIND ATTACK AND EVALUATION

Attacker observes protected records, not keys or slot labels; slots are revealed only in a labelled control.



KEY DESIGN Fresh: source-record keys are split-disjoint. Pool K: hidden transforms recur across identity splits.

REPLICATION MOBIO / FEI original; MOBIO / SCface extended: 2 splits x 3 seeds; 120-epoch caps.

DESIGN Three-pool prediction-mean baseline; raw learned and stricter native studies are separate controls.

System overview: fixed encoder, hidden-key protection, same-identity exposure sets and held-out gallery linkage.

Detailed attacker: aggregation and reconstruction

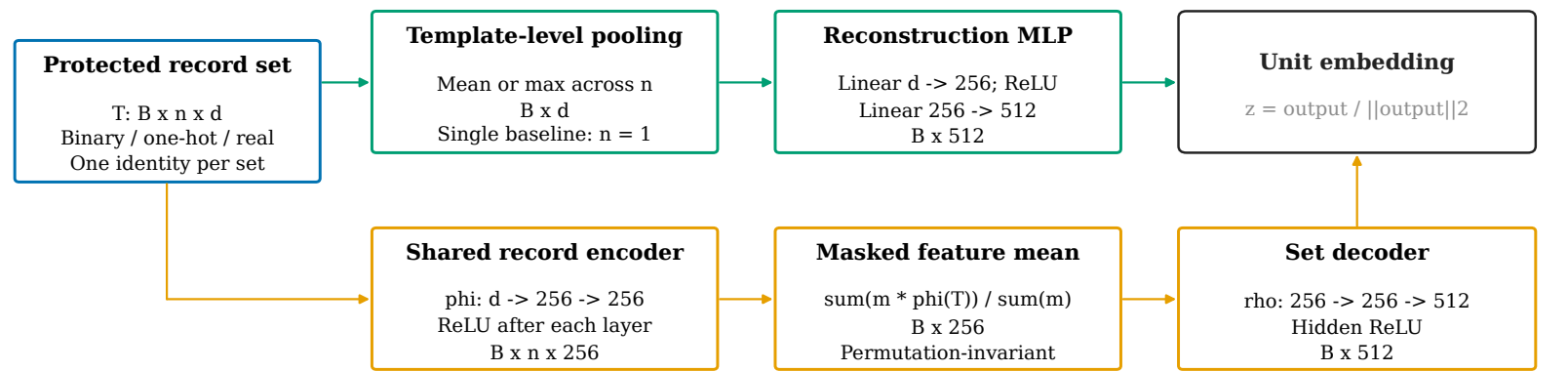
A MATCHED EXPOSURE CONSTRUCTION

One identity, n source records; nested subsets from one fixed permutation per repeat; gallery excluded.

Fresh: independent source-record keys. Pool K: hidden transforms recur across identity splits. No keys or slots in model input.

T: $B \times n \times d$ $d = 128$ BioHash; 512 MLP-Hash; 4,800 one-hot IoM-GRP; 170 PolyProtect

B TWO AGGREGATION PATHS



Primary: mean-pool MLP. Secondary: DeepSets. m masks valid records; all n records are valid in these runs.

C SUPERVISION

Target $u = \text{normalize}(\text{mean of exposed source embeddings})$.

Loss = $\text{mean}[1 - \cos(z, u)] + 0.1 \times \text{MSE}(z, u)$.

Adam: lr 0.001; weight decay 0.0001; hidden width 256.

Train identities only; select minimum validation loss; freeze test.

D HELD-OUT IDENTITY LINKAGE

Scores = $z \cdot G^T$; G: $N \times 512$ unprotected unit gallery.

Rank identities; top-1/top-5, AUROC, EER, TAR at FAR.

Follow-up: 3 model seeds x 2 identity partitions; 120 epochs.

Crossed seed/identity intervals; paired primary contrasts.

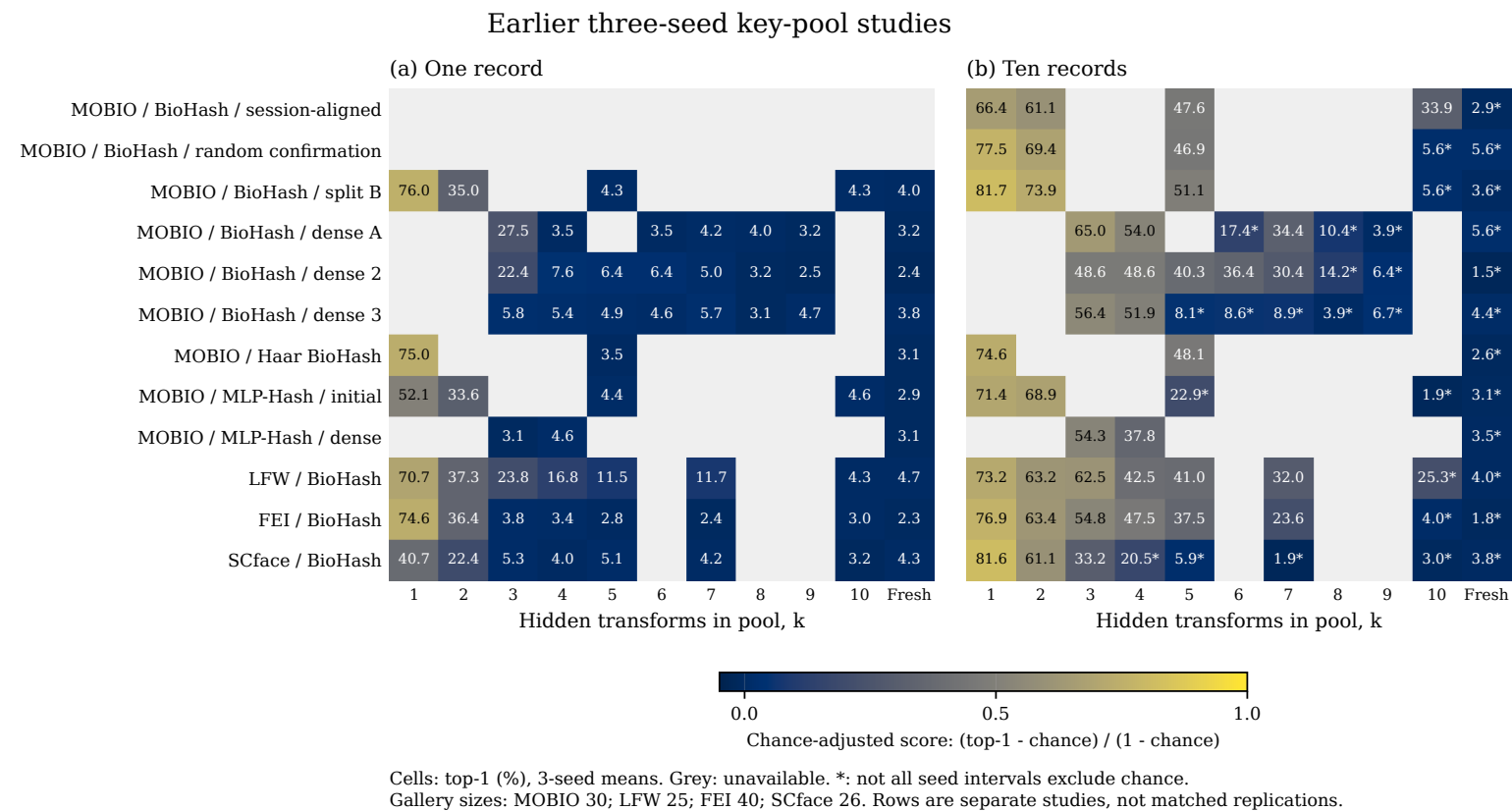
B: batch size; n: exposure count; d: template input width; N: test gallery identities. Source targets are not inference inputs.

Mean pooling is the primary endpoint; DeepSets is secondary. Target is the normalized mean of exposed embeddings, not the gallery.

Cross-dataset key-reuse boundary

SCface: pool 3 rises from 5.29% to 33.17% top-1; fresh keys give 3.85%.

SCface pools 1/2/3 pass the all-seed interval criterion; 4/5/7/10 fail. Failure is not equivalence.

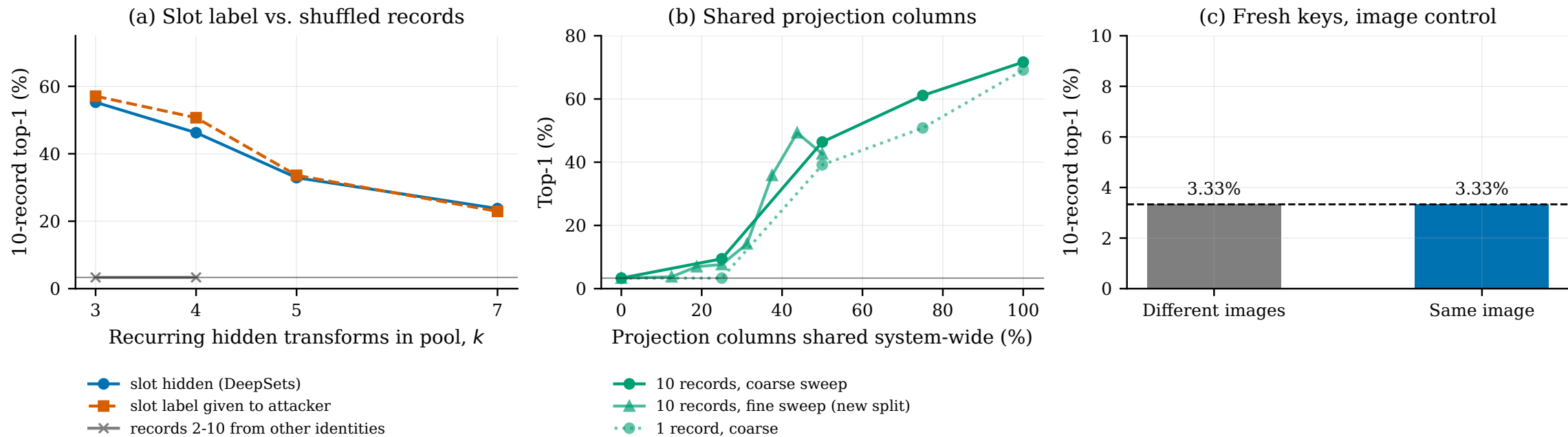


73 conditions / 12 source-separated, three-seed studies. Grey: unavailable. *: interval failure. Pilots excluded.

Mechanism controls and interpretation

MOBIO: slot labels change DeepSets by -0.83 to +4.44 points; shuffled records collapse to 3.33% chance.
Same-image fresh-key sets also give 3.33%. Projection-sharing effects depend on construction and partition.

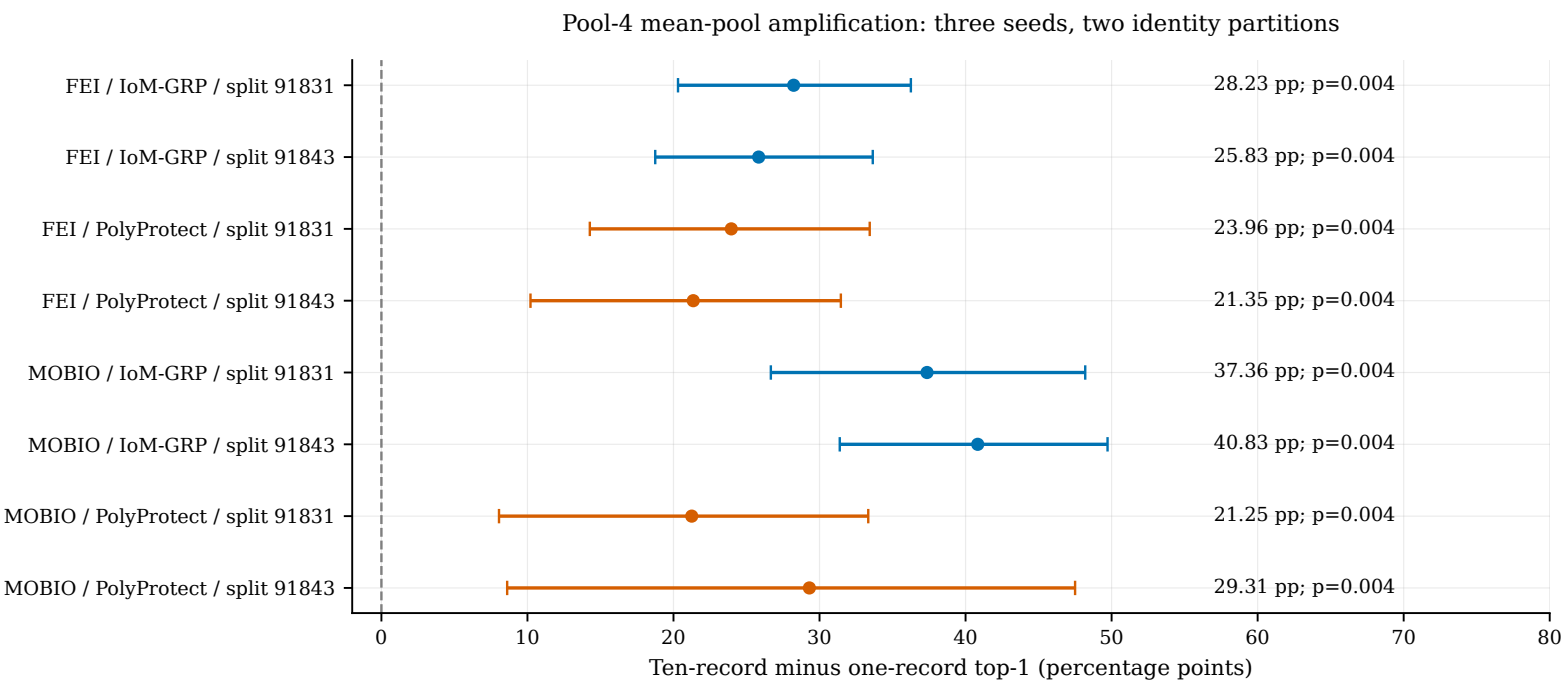
MOBIO mechanism controls (30 gallery identities; chance 3.33%)



Sources: experiments/mobio_mechanism_controls and mobio_correlation_controls. Three model seeds; chance 1/30.

Multi-seed, multi-partition validation

216 trained endpoints; fresh, shared and pool-4 keys; 3 model seeds x 2 identity partitions.
All 8 planned pool-4 amplification tests pass Holm correction (maximum adjusted $p = 0.004$).

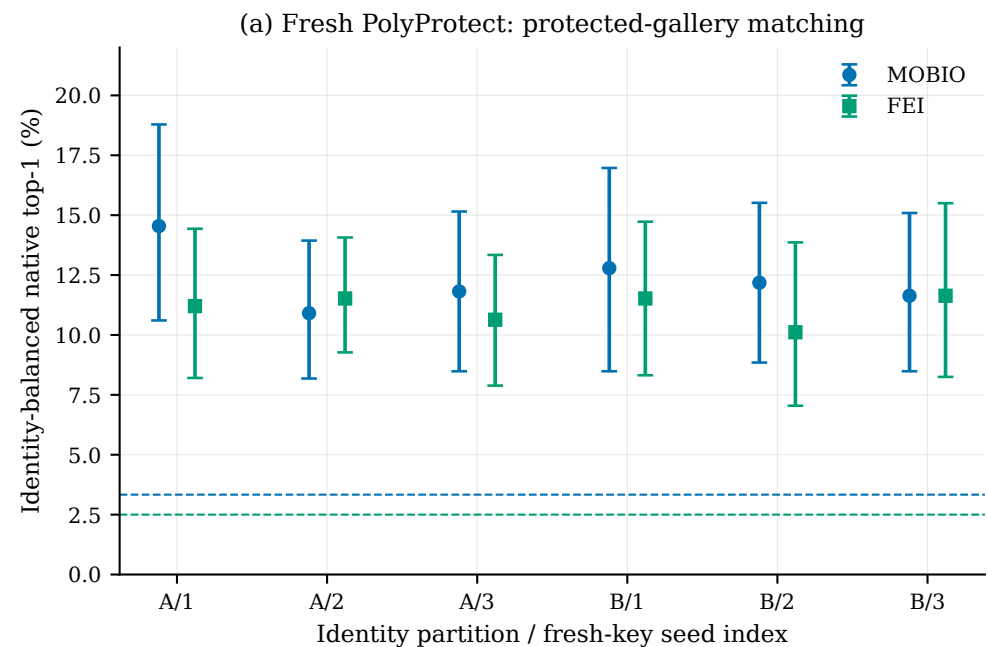


Bars: crossed model-seed / identity bootstrap 95% intervals. Tests: identity sign-flips, Holm family of 8.
Matched 120-epoch caps; key/set seeds fixed. Partitions are separate sensitivity analyses, not independent replications.

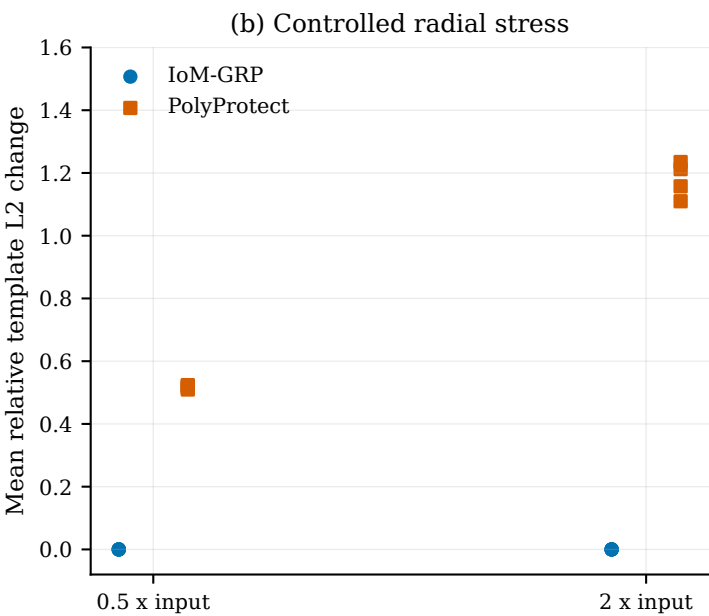
Matched 120-epoch caps. Crossed-bootstrap intervals include model-seed and identity resampling within each partition.

Native linkage and radial sensitivity

Fresh PolyProtect native identification exceeds the gallery-label null in 12 / 12 tests.
Maximum Holm-adjusted $p = 0.006$; three fresh-key seeds per dataset and identity partition.



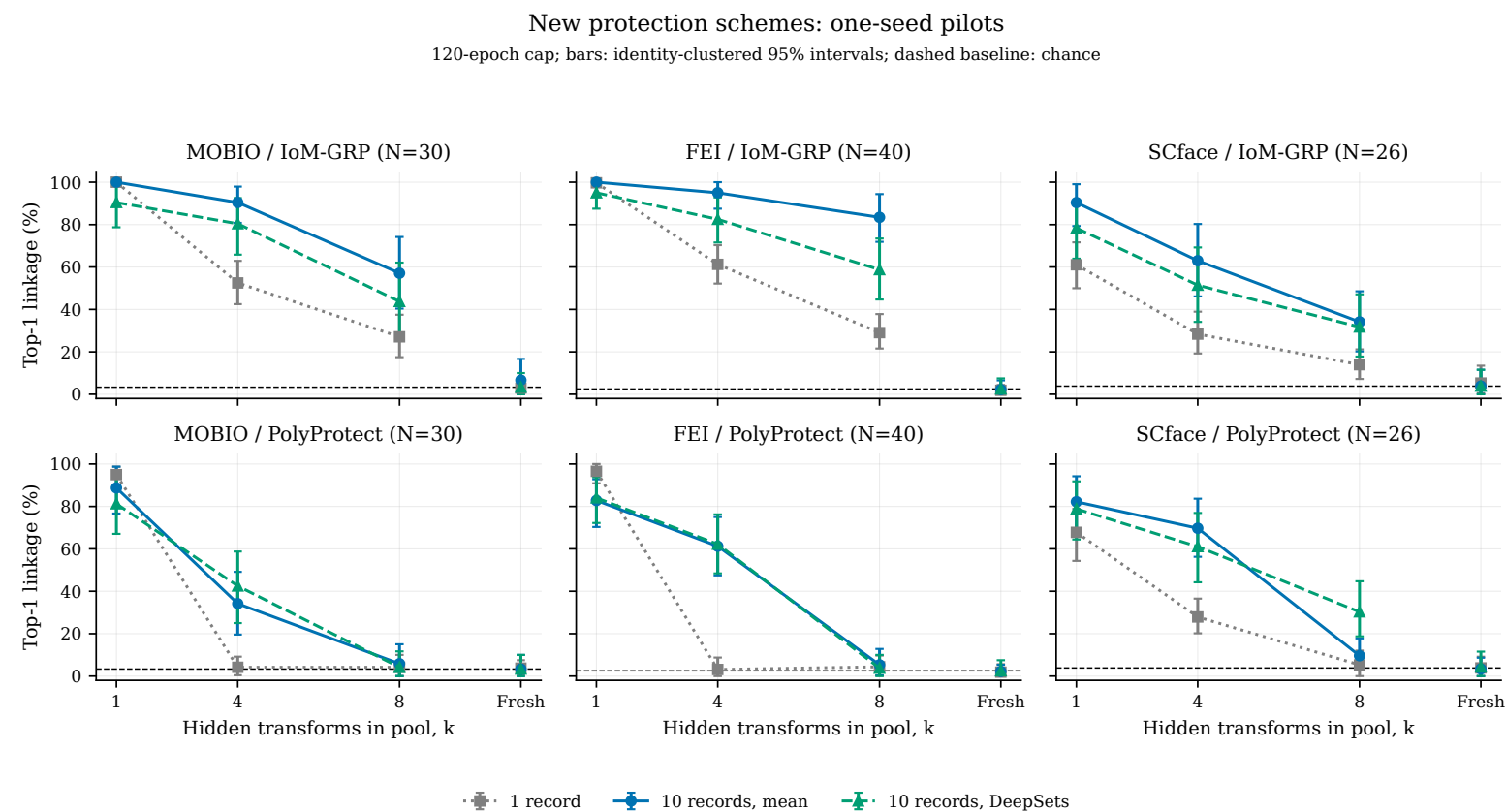
Native tests: 12 / 12 Holm-adjusted $p \leq 0.006$; identity-clustered 95% intervals; dashed lines: chance.
A/B: splits 91831/91843. Key seeds 1/2/3: 91873/91879/91883. Gallery-label permutations preserve probe clusters.
Radial stress: 32 held-out records per dataset/partition, fixed keys; measures scale sensitivity, not natural norm leakage.



Earlier synthetic stress: IoM-GRP unchanged, PolyProtect scale-sensitive. Genuine raw-input controls follow on page 12.

Earlier cross-dataset scheme pilots

Earlier one-seed pilots: 72 model endpoints on MOBIO / FEI / SCface, including pool 8.
Separate native fresh-key PolyProtect diagnostic: 12.73% / 13.73% / 10.66% top-1, respectively.



Pilot intervals condition on one seed and partition; not confirmation. New replicated MOBIO/FEI results are on page 6.

Closest research and our precise contribution

WHAT PRIOR WORK ALREADY SHOWED

Record-multiplicity attacks: several templates plus known parameters can enable recovery.

PolyProtect (2022): 1-10-record inversion; residual linkage under naive random parameters; stricter parameter selection improves unlinkability. Our native result is not a first discovery.

benchmark_cb: recognition, score-based unlinkability and information estimates across schemes.

FaceLinkGen v3: adaptive identity distillation from paired data with hidden per-query randomness.

Maximal-linkability work: joint similarity scores, multiple keys and composition limits (thesis 5.4).

WHAT OUR CONTROLLED EXPERIMENT ADDS

Different-image set aggregation with hidden fresh/shared/pool keys and disjoint training identities.

A measured one-to-ten benefit conditional on transform reuse, plus mechanism and failure controls.

Independent pool draws and prediction averaging expose pool sensitivity and a competitive baseline.

NOT CLAIMED

Multiplicity, identity distillation and IoM scale invariance are not new.

No head-to-head accuracy win, exhaustive priority claim or break of stricter PolyProtect policies.

Full-text evidence: author thesis 5.4, linked to Access 2024; publisher PDF blocked. See docs/literature/closest_work_2026-09-18.md.

Attacker access: assumptions and limits

SAME-PERSON RECORDS

An account or session pseudonym could group retained records without revealing real identity.
Stable grouping and retention are assumed; their prevalence in deployed products is not measured.

PAIRED TRAINING EXAMPLES

An authorized enrollment/query interface or a provider seeing inputs and outputs could supply pairs.
Crucially, training must use the SAME realized hidden pool as the target records.
A proxy with unrelated keys is insufficient. Identity splits are disjoint; recurring keys are not.

REFERENCE GALLERY

Lawful public or consented reference images; gallery images are excluded from exposure sets.
Only 25-40 candidate people, with guaranteed target membership: a small closed-set experiment.

BOUNDARIES

No open-set search, internet-scale distractors, unknown grouping or cross-provider transfer test.
Fresh independent transforms can also harm legitimate matching; they are not a tested defense.

Separate implementation and matching checks

FORMULA AND PARAMETER CHECKS

PolyProtect: separate scalar windows, zero padding, coefficients and powers match local outputs.

IoM-GRP: explicit grouped dot-product/argmax reference agrees; paper dimensions are tested.

Original PolyProtect uses different encoders and offers a stricter parameter-selection policy.

NATIVE MATCHING AUDIT: 48 MATCHED CELLS

Scalar PolyProtect and production outputs: zero difference after float32 casting.

Matrix and SciPy cosine scores: maximum difference $1.33\text{e-}15$; identical predictions.

Zero gallery/probe overlap, score ties, reversed-gallery changes or duplicate fresh keys.

The unit-input results reproduce the earlier three-key native averages.

RAW EXTRACTION CHECK

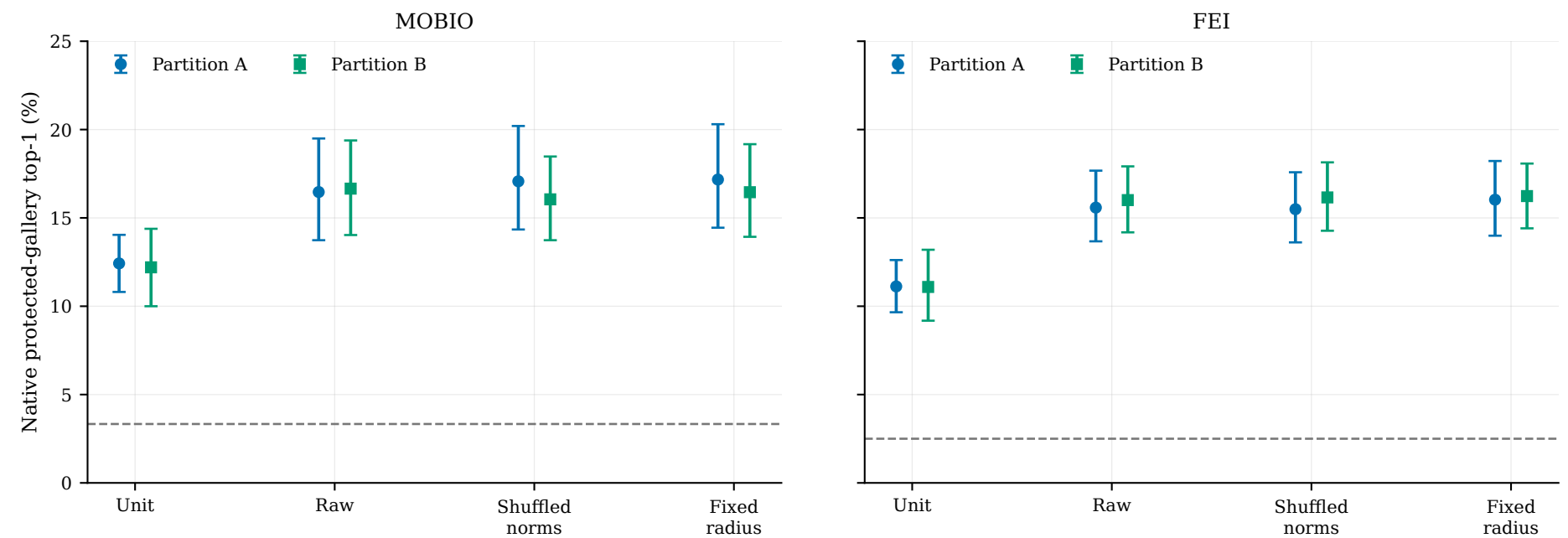
All 4,177 MOBIO/FEI records re-extracted; normalized discrepancy at most $2.98\text{e-}8$.

Natural raw/unit scaling changes zero of 38,400 sampled IoM codes.

Separate computational checks, not independent human review or official-code certification.

Raw embeddings: scale is not identity leakage

Real raw embeddings, shuffled norms and a fixed training-median radius; same keys in every arm.
Raw matching rises to 15.58-16.66%, but shuffled norms perform similarly; identity-specific norm leakage is not established.

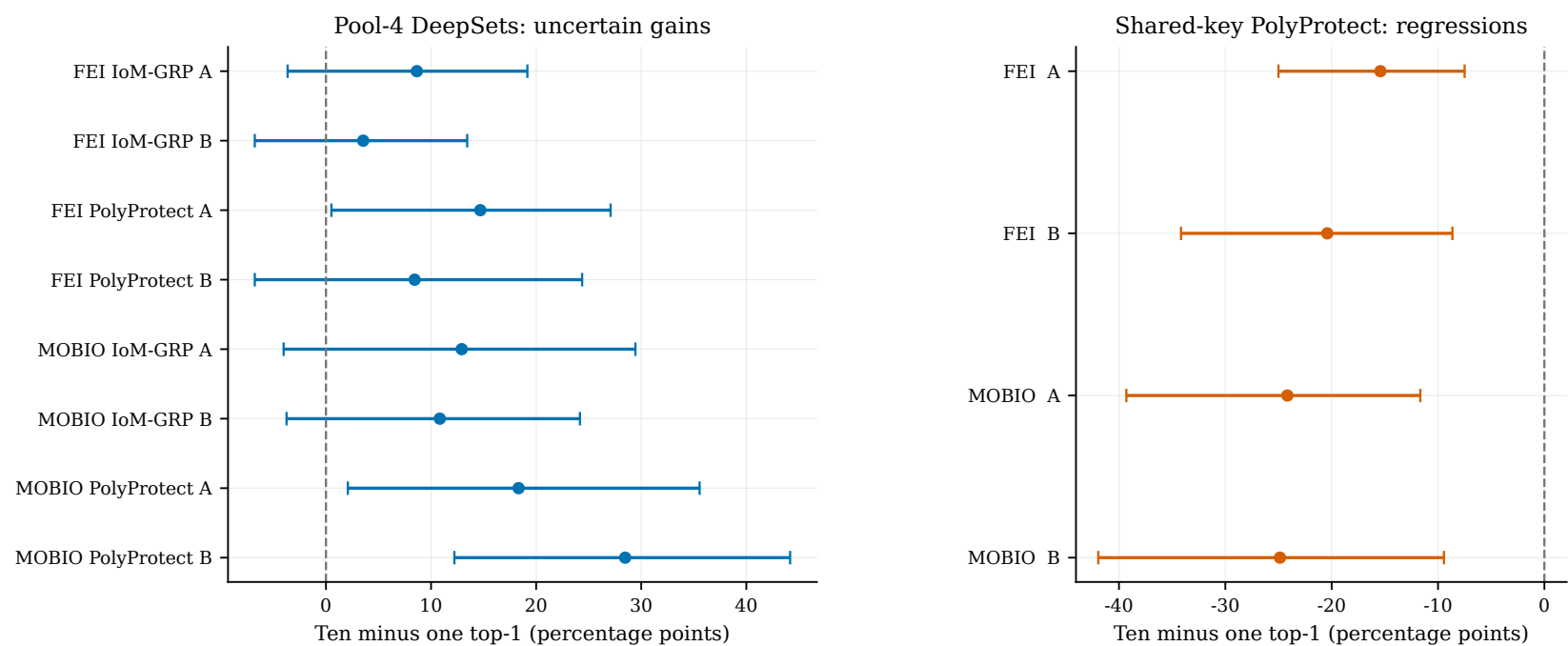


Three fixed key seeds averaged; identity-bootstrap 95% intervals. All 16 null tests: Holm $p = 0.0032$.
Raw minus unit survives paired correction only on FEI. Raw minus shuffled: all adjusted $p \geq 0.7584$.
Natural identity-specific norm leakage is not established. No learned raw-input attack was retrained.

Raw-unit gain survives only on FEI. This page is native matching; separate learned raw-input results are on page 18.

Failure analysis: all 48 contrasts

Post-hoc two-sided family of 48; original primary eight-test family remains unchanged.
Eight pool-4 mean gains survive; four shared-key PolyProtect DeepSets losses also survive (adjusted $p = 0.0192$).



All 48 contrasts reported; post-hoc two-sided Holm family 48. Pointwise crossed seed/identity 95% intervals.
Left: none survives correction (minimum $p = 0.0576$). Right: all four negative contrasts survive ($p = 0.0192$).
All eight pool-4 mean-pool gains remain significant ($p = 0.0192$). More records do not universally improve linkage.

Seed ranges and leave-one-seed-out gains are published. Negative results constrain the claim: more records are not always better.

Findings and study scope

TRANSFORM REUSE AND RECORD MULTIPLICITY

Recurring hidden transforms support substantial identity linkage from multiple protected records. The transition depends on dataset, partition and realized pool, not a universal pool-size threshold. SCface pool 3: 5.29% single-record versus 33.17% ten-record top-1; chance is 3.846%.

BOUNDARIES AND FAILURES

Shuffled-identity controls constrain the mechanism; same-pool training access remains essential. Shared-key PolyProtect DeepSets loses 15.42-24.86 points: aggregation is not universally beneficial. Native matching survives separate checks, but natural identity-specific norm leakage is not established.

STATISTICAL AND THREAT-MODEL SCOPE

BioHash spans four datasets; scheme follow-ups cover MOBIO/FEI and now MOBIO/SCface. New scheme follow-up: 216 endpoints; all 8 primary contrasts pass Holm correction. New three-pool study: IoM gains persist; PolyProtect is sensitive to the pool draw. All direct prediction-mean baseline intervals include zero: no input-pooling superiority claim. Fresh-key learned intervals include chance; they do not establish equivalence or general unlinkability.

Scope: only three new pools, fixed set seed, overlapping assignments, one encoder, small galleries, paired access. Three extended-study source snapshots remain unresolved locally; independent human review remains open.

Four datasets: coverage and evidence strength

All four datasets are shown, without mixing historical studies, one-seed pilots and matched follow-ups.
SCface now has extended training and controls; LFW and FEI do not have the same complete new matrix.

Dataset coverage: evidence layers are not interchangeable

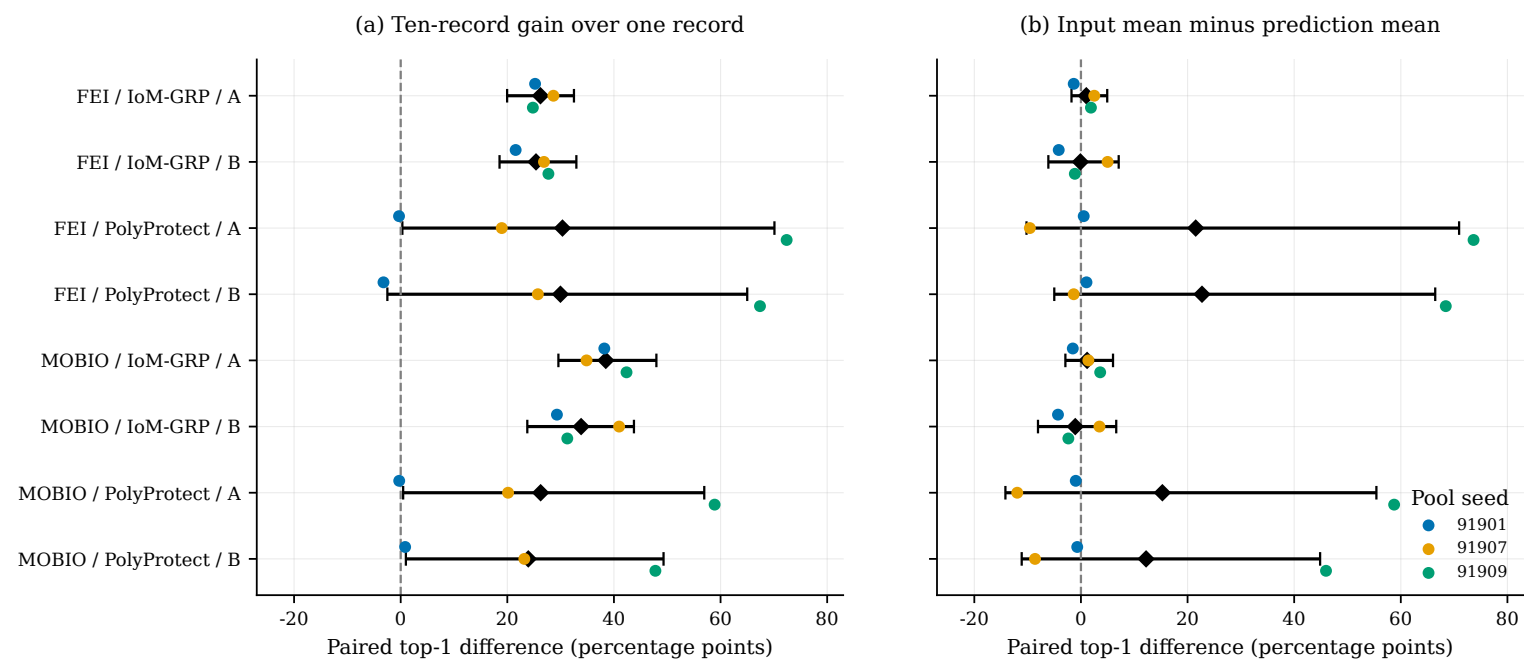
BioHash historical	3 model seeds	3 model seeds	3 model seeds	3 model seeds
MLP-Hash historical	3 model seeds	Not run	Not run	Not run
IoM / PolyProtect pilots	1 model seed	Not run	1 model seed	1 model seed
Original scheme follow-up	3 seeds / 2 splits	Not run	3 seeds / 2 splits	Not run
Native / raw-norm audit	3 keys / 2 splits	Not run	3 keys / 2 splits	Not run
Independent-pool + baseline	3 pools / 3 seeds 2 splits	Not run	3 pools / 3 seeds 2 splits	Not run
Extended exposure study	3 seeds / 2 splits	Not run	Not run	3 seeds / 2 splits
Learned raw-input study	3 seeds / 1 split	Not run	Not run	3 seeds / 1 split
Stricter-selection audit	3 keys / 1 split	Not run	Not run	3 keys / 1 split
	MOBIO	LFW	FEI	SCface

SCface now has an extended study; coverage alone does not establish identical gallery protocols.
Grey = not run in this package, not zero accuracy. Identity partitions overlap; native audits do not train attackers.

Evidence coverage, not a performance ranking. LFW remains historical BioHash evidence; missing studies require experiments.

Independent pool draws and matched baseline

24 cells; 3 new pool draws x 3 model seeds x 2 partitions; 144 fits plus 72 prediction-mean endpoints.
Matched gallery and records; prediction mean reuses single-record training, not the set-training objective.
Pool seeds vary transforms and slot assignments jointly. Intervals are pointwise; only three pool draws.



Colored dots: individual pool draws, averaged over three model seeds. Black: mean and crossed pool/seed/identity 95% interval.

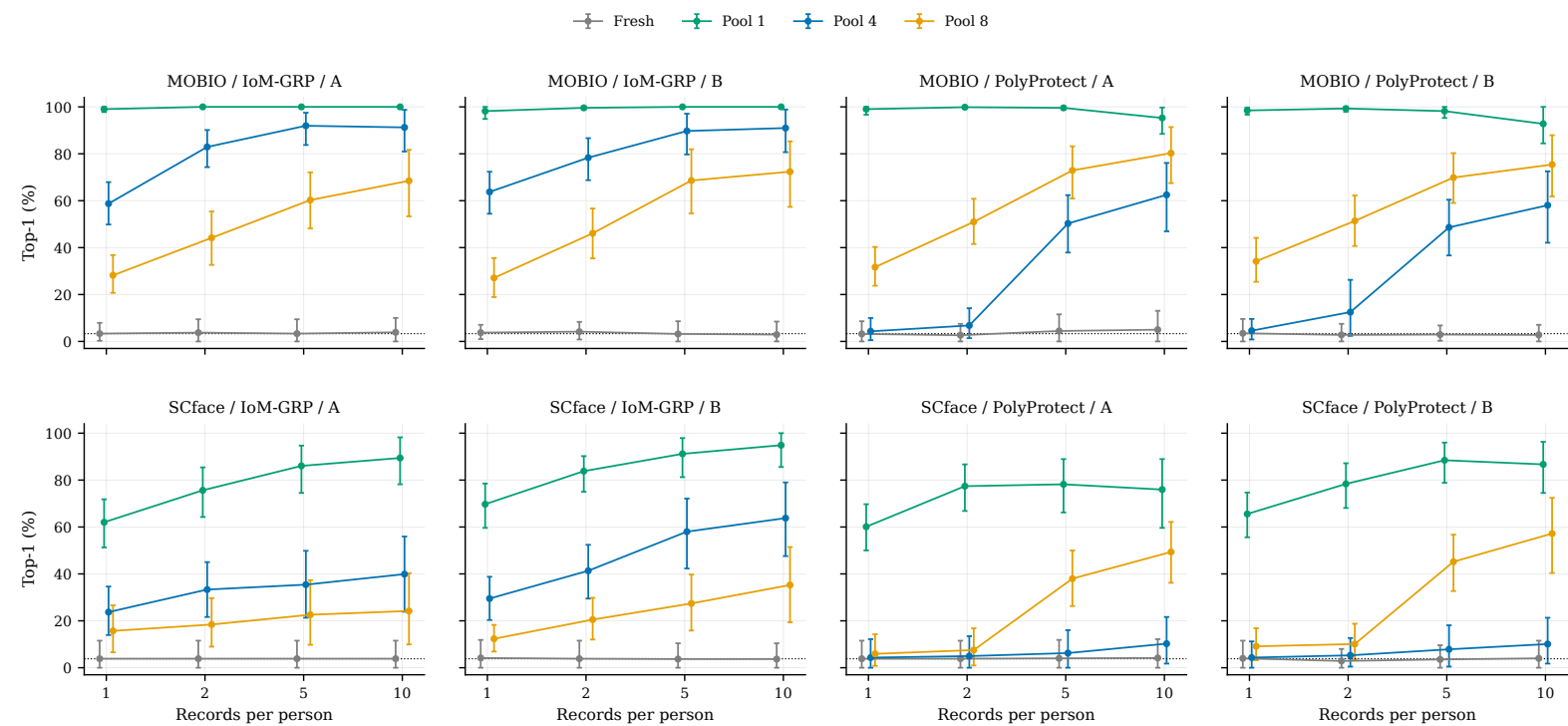
IoM gains persist across draws; PolyProtect is pool-sensitive. Every direct baseline-comparison interval includes zero.

Three pools only; pointwise uncertainty, no corrected significance claim. A/B: overlapping partitions 91831/91843.

IoM per-pool gains: +21.56 to +42.36 pp. PolyProtect: -3.23 to +72.40 pp. Three pools do not establish universal robustness.

Extended exposures: MOBIO and SCface

32 cells / 672 endpoints: 3 seeds x 2 splits; exposures 1/2/5/10; fresh and pools 1/4/8.
Eight pool-4 primary gains pass Holm correction ($p = 0.004\text{-}0.034$); all 56 fresh endpoint intervals include chance.

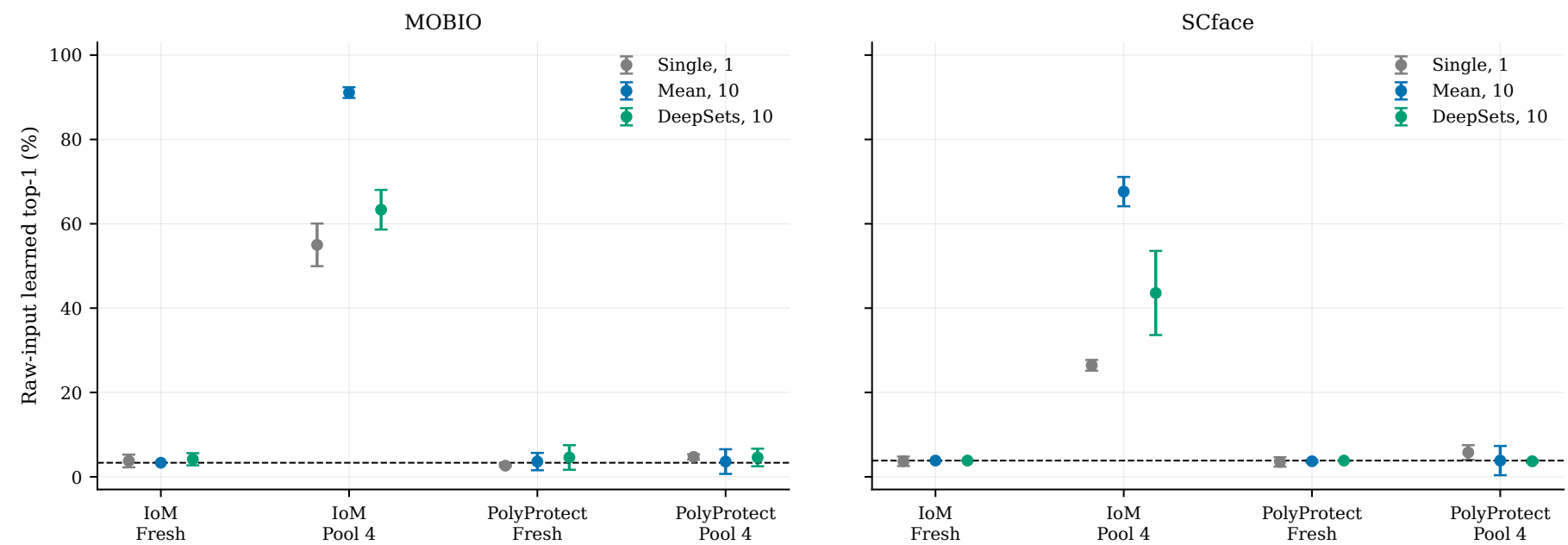


Single MLP at 1 record; mean MLP at 2/5/10. Bars: crossed seed/identity 95% intervals; dotted line: chance.
A/B are overlapping identity assignments. One pool realization per condition; pool size is not an isolated causal effect.

SCface pool-4 gains: IoM +16.19/+34.29 pp; PolyProtect +5.93/+5.77 pp. One pool-8 draw cannot isolate a pool-size effect.

Learned raw-input attacks: a separate study

24 summaries / 72 fits; one saved partition; three model seeds; raw embeddings before protection.
Pool-4 mean top-1: PolyProtect 3.61% / 3.85%; IoM 91.11% / 67.63% on MOBIO / SCface.

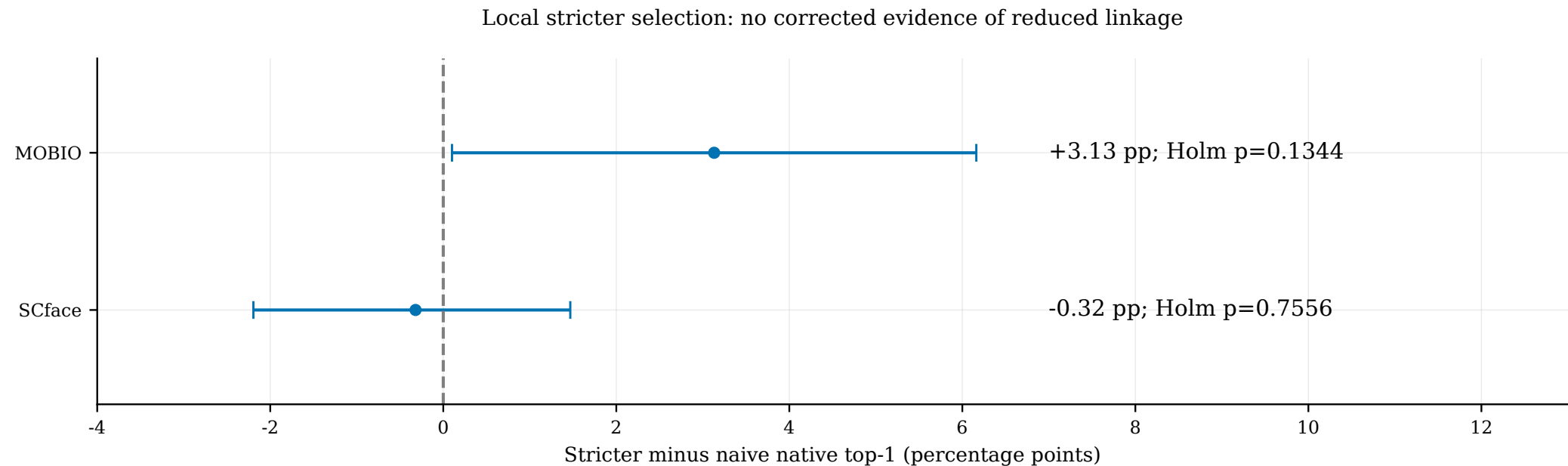


Bars: standard deviation across three model seeds, NOT confidence intervals. One saved identity split; descriptive study.
PolyProtect is near chance in this run; IoM retains large pool-4 gains. No matched unit-input retraining at these seeds.
Different pool/set seeds and partitions prevent attributing cross-study accuracy differences solely to normalization.

Descriptive, not a corrected null test. Raw inputs also change set targets; normalization alone is not isolated. No privacy claim.

Local stricter PolyProtect selection

Native protected-gallery matching, not learned reconstruction: three fixed key seeds per dataset.
MOBIO: 13.13% -> 16.26%; SCface: 10.44% -> 10.12%. Neither paired change survives correction.



Left of zero would mean less native linkage. Bars: identity-bootstrap 95% intervals, conditional on three key seeds.
20 candidates; 200 development pairs; score band [-0.5, 0.5]. Paired Holm family 2; neither comparison passes 0.05.
Local operationalization, not source-exact replication or a refutation of all stricter PolyProtect parameter policies.

All four arms exceed the gallery-label null (Holm $p = 0.0008$). This does not certify or invalidate every stricter selection policy.